VRUSHABH SAWANT SYVLSI_44 125BTVL1033

Experiment 6

Academic Year 2026-2027			
Program: Electronics Engineering (VLSI Design and Technology)			
Year of Study	Second Year	**Semester**	III
Course	Skill Lab: Python Programming	**Course Code**	VDCOR2VS201
Class	SY VLSI	**Course In-Charge**	Mr. Nirmol Munvar
Exp Name	Functions, default arguments, lambda functions		
CO	Analyze and implement control structures and user-defined functions to solve logical and numerical problems using Python.		

Functions

A **function** is a reusable block of code designed to perform a specific task. Functions help make programs modular, reduce code repetition, and improve readability.

A function is defined using the def keyword.

Basic syntax:

```
def function_name(parameters):
    statements
```

The function is executed by calling its name.

```
def greet():
    print("Hello")

greet()
```

Function with Parameters

Parameters allow data to be passed into a function.

```
def add(a, b):
    print(a + b)

add(10, 20)
```

Here, a and b are parameters, while 10 and 20 are arguments passed during the function call.

Function with Return Value

A function can return a result using the return statement.

```
def add(a, b):
    return a + b

result = add(10, 20)
print(result)
```

return sends the result back to the point where the function was called.

A function can have multiple parameters and can also return multiple values.

```
def calculate(a, b):
    return a + b, a - b

sum_value, difference = calculate(20, 10)
print(sum_value)
```

```
print(difference)
```

Default Arguments

A **default argument** is a parameter that has a predefined value. If the caller does not provide a value for that parameter, Python uses the default value.

Syntax:

```
def function_name(parameter=default_value):
    statements
```

Example:

```
def greet(name="Student"):
    print("Hello", name)

greet("Rahul")
greet()
```

Output:

```
Hello Rahul
Hello Student
```

Default arguments are useful when a parameter usually has a standard value but can be changed when required.

A function can have multiple default arguments.

```
def student(name, branch="VLSI", year=3):
    print(name, branch, year)

student("Rahul")
student("Priya", "ECE", 4)
```

Parameters with default values should generally be placed after parameters without default values.

```
def student(name, branch="VLSI"):
    print(name, branch)
```

Lambda Functions

A **lambda function** is a small anonymous function. It is useful when a simple function is required for a short operation.

Syntax:

```
lambda arguments: expression
```

Unlike a normal function defined using def, a lambda function normally contains a single expression.

Example:

```
square = lambda x: x * x

print(square(5))
```

Output:

```
25
```

Another example:

```
add = lambda a, b: a + b

print(add(10, 20))
```

Lambda functions are commonly used with functions such as map(), filter(), and sorted().

For example:

```
numbers = [1, 2, 3, 4, 5]

squares = list(map(lambda x: x * x, numbers))
```

print(squares)
A lambda function can also contain a conditional expression.
check = lambda x: "Even" if x % 2 == 0 else "Odd"

print(check(7))
For larger or more complex operations, a regular def function is generally more readable than a lambda function.

Basic Exercises

Exercise 1: Basic Functions

Write a Python program containing separate functions to:

- Calculate the sum of two numbers.
- Calculate the difference between two numbers.
- Calculate the product of two numbers.
- Calculate the division of two numbers.

Accept the values from the user and call the appropriate functions to display the results.

Exercise 2: Default Arguments

Write a function student_details() that accepts:

- Student name
- Branch, with "VLSI" as the default value
- Year, with 3 as the default value

Call the function using different combinations of arguments and display the student details.

Exercise 3: Lambda Functions

Given a list of numbers, use lambda functions to:

- Find the square of each number.
- Check whether each number is even or odd.
- Find the cube of each number.

Use map() where appropriate and display the resulting lists.

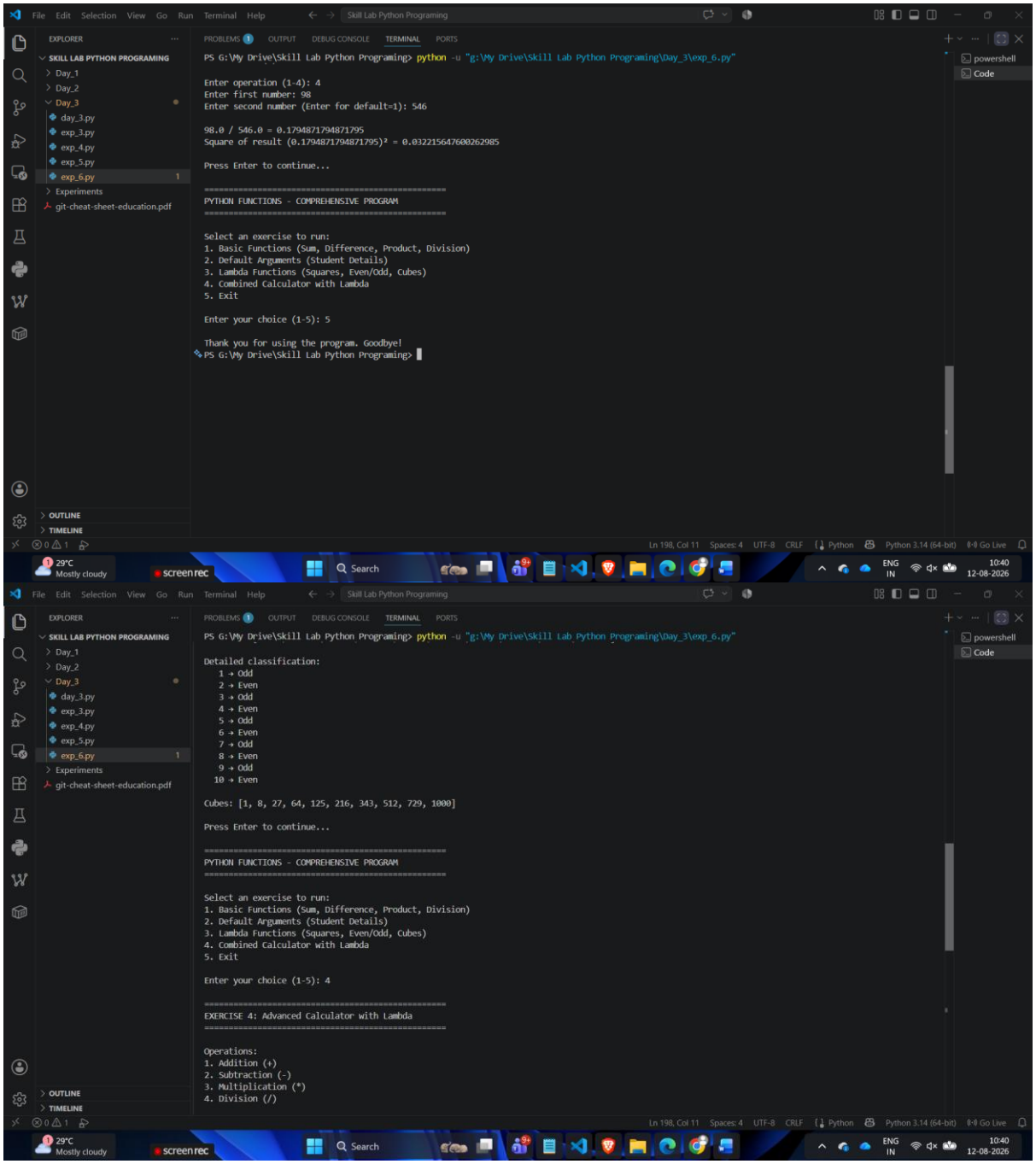
Exercise 4: Combined Function Exercise

Write a simple calculator using functions. Create separate functions for addition, subtraction, multiplication, and division.

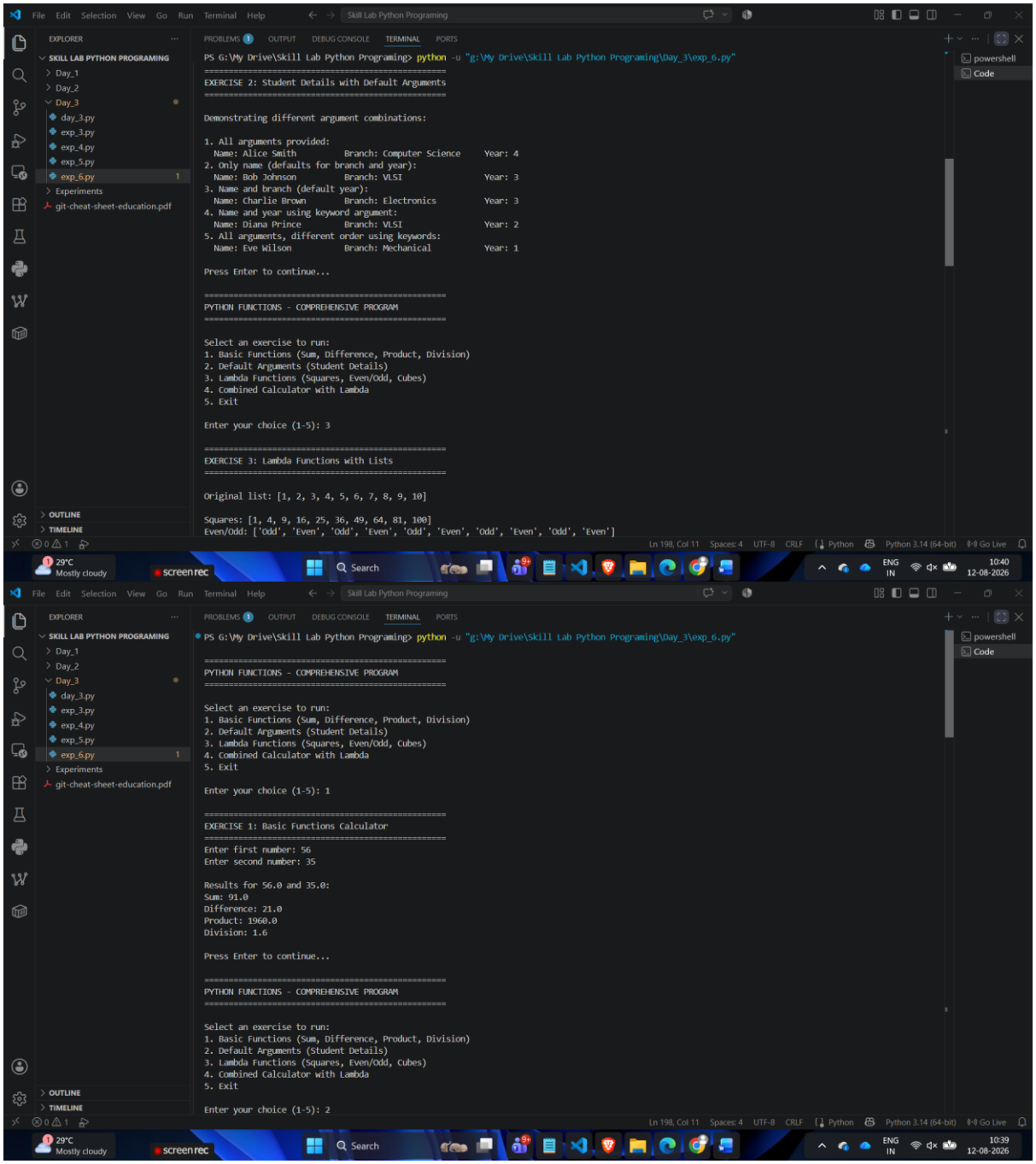
Use **default arguments** for the division function to provide a default divisor of 1.

Use a **lambda function** to calculate the square of the final result.

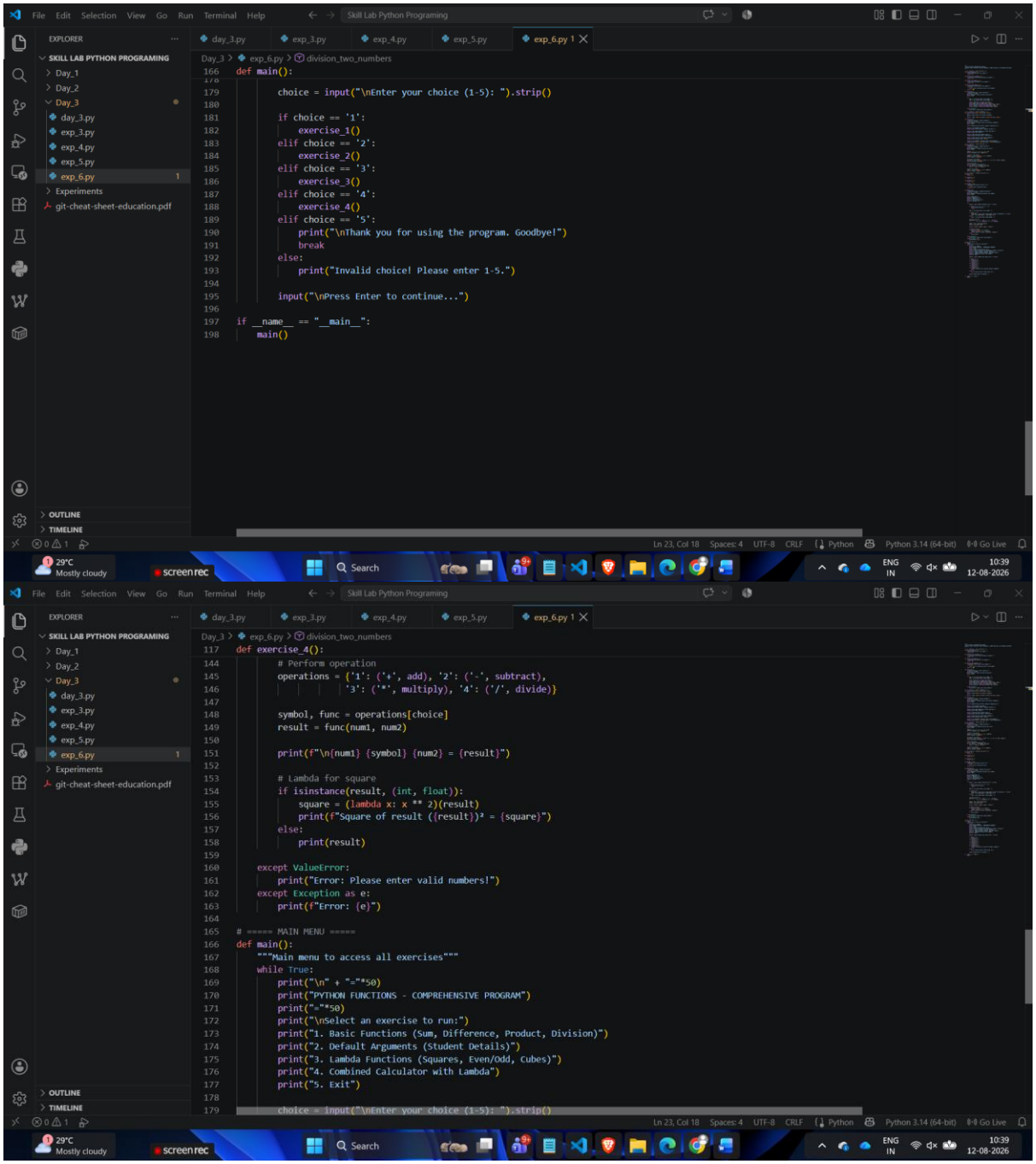
For example, if the user enters 10 and 5 for addition, the program should calculate 15 and then use the lambda function to calculate its square, 225.


File Edit Selection View Go Run Terminal Help
Skill Lab Python Programing
EXPLORER
SKILL LAB PYTHON PROGRAMING
Day_1
Day_2
Day_3
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py
Experiments
git-cheat-sheet-education.pdf
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS G:\My Drive\Skill Lab Python Programing> python -u "g:\My Drive\Skill Lab Python Programing\Day_3\exp_6.py"
Enter operation (1-4): 4
Enter first number: 98
Enter second number (Enter for default=1): 546
98.0 / 546.0 = 0.1794871794871795
Square of result (0.1794871794871795)² = 0.032215647600262985
Press Enter to continue...
PYTHON FUNCTIONS - COMPREHENSIVE PROGRAM
Select an exercise to run:
1. Basic Functions (Sum, Difference, Product, Division)
2. Default Arguments (Student Details)
3. Lambda Functions (Squares, Even/Odd, Cubes)
4. Combined Calculator with Lambda
5. Exit
Enter your choice (1-5): 5
Thank you for using the program. Goodbye!
PS G:\My Drive\Skill Lab Python Programing>
powershell
Code
OUTLINE
TIMELINE
Ln 198, Col 11 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Go Live
29°C
Mostly cloudy
screenrec
Search
ENG
IN
10:40
12-08-2026

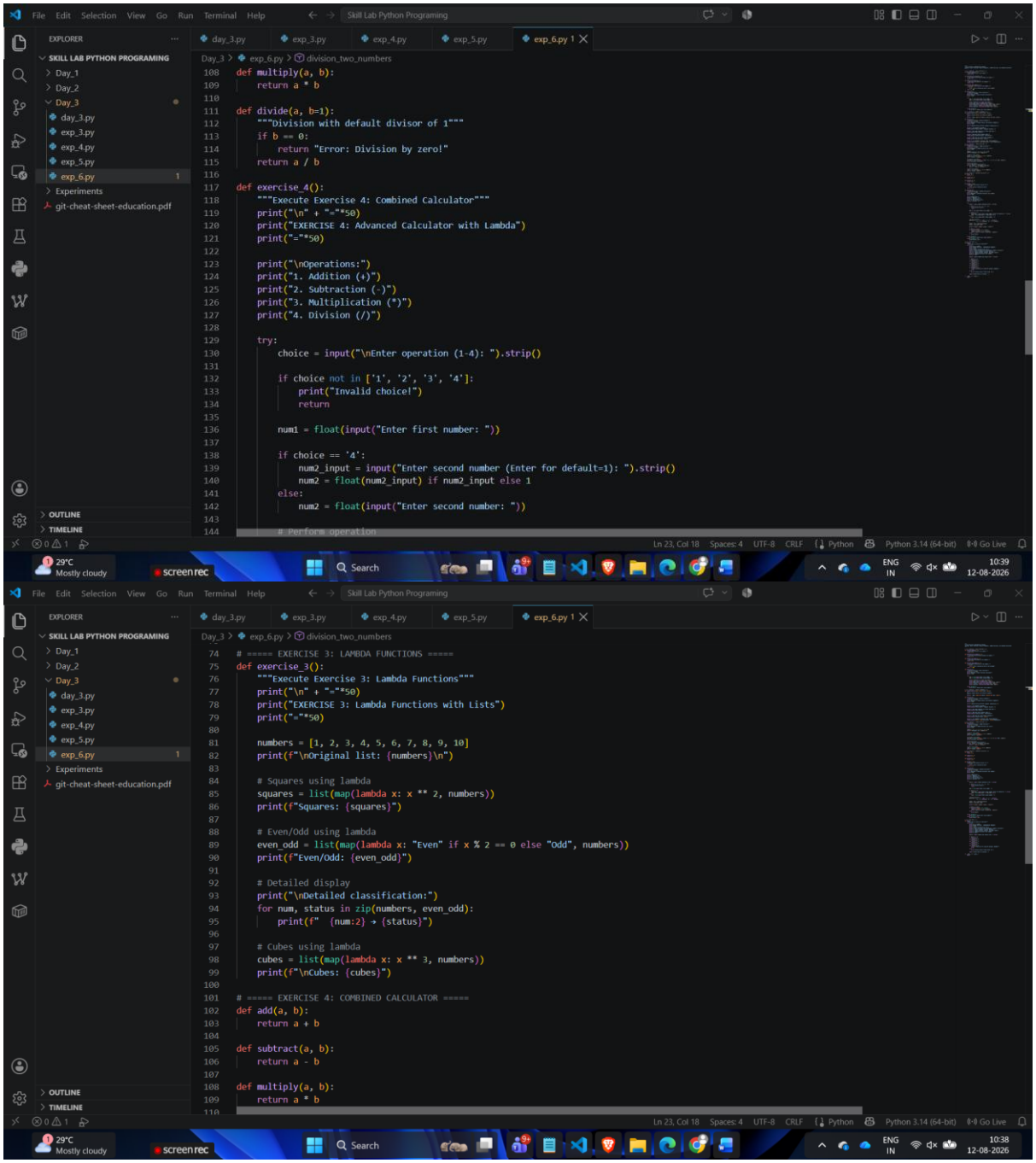
File Edit Selection View Go Run Terminal Help
Skill Lab Python Programing
EXPLORER
SKILL LAB PYTHON PROGRAMING
Day_1
Day_2
Day_3
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py
Experiments
git-cheat-sheet-education.pdf
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS G:\My Drive\Skill Lab Python Programing> python -u "g:\My Drive\Skill Lab Python Programing\Day_3\exp_6.py"
Detailed classification:
1 → Odd
2 → Even
3 → Odd
4 → Even
5 → Odd
6 → Even
7 → Odd
8 → Even
9 → Odd
10 → Even
Cubes: [1, 8, 27, 64, 125, 216, 343, 512, 729, 1000]
Press Enter to continue...
PYTHON FUNCTIONS - COMPREHENSIVE PROGRAM
Select an exercise to run:
1. Basic Functions (Sum, Difference, Product, Division)
2. Default Arguments (Student Details)
3. Lambda Functions (Squares, Even/Odd, Cubes)
4. Combined Calculator with Lambda
5. Exit
Enter your choice (1-5): 4
EXERCISE 4: Advanced Calculator with Lambda
Operations:
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
powershell
Code
OUTLINE
TIMELINE
Ln 198, Col 11 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Go Live
29°C
Mostly cloudy
screenrec
Search
ENG
IN
10:40
12-08-2026


File Edit Selection View Go Run Terminal Help
Skill Lab Python Programing
EXPLORER
SKILL LAB PYTHON PROGRAMING
Day_1
Day_2
Day_3
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py
Experiments
git-cheat-sheet-education.pdf
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS G:\My Drive\Skill Lab Python Programing> python -u "g:\My Drive\Skill Lab Python Programing\Day_3\exp_6.py"
EXERCISE 2: Student Details with Default Arguments
Demonstrating different argument combinations:
1. All arguments provided:
Name: Alice Smith Branch: Computer Science Year: 4
2. Only name (defaults for branch and year):
Name: Bob Johnson Branch: VLSI Year: 3
3. Name and branch (default year):
Name: Charlie Brown Branch: Electronics Year: 3
4. Name and year using keyword argument:
Name: Diana Prince Branch: VLSI Year: 2
5. All arguments, different order using keywords:
Name: Eve Wilson Branch: Mechanical Year: 1
Press Enter to continue...
PYTHON FUNCTIONS - COMPREHENSIVE PROGRAM
Select an exercise to run:
1. Basic Functions (Sum, Difference, Product, Division)
2. Default Arguments (Student Details)
3. Lambda Functions (Squares, Even/Odd, Cubes)
4. Combined Calculator with Lambda
5. Exit
Enter your choice (1-5): 3
EXERCISE 3: Lambda Functions with Lists
Original list: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Squares: [1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
Even/Odd: ['Odd', 'Even', 'Odd', 'Even', 'Odd', 'Even', 'Odd', 'Even', 'Odd', 'Even']
OUTLINE
TIMELINE
Ln 198, Col 11 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Go Live
powershell
Code
29°C Mostly cloudy
screenrec
Search
ENG IN
10:40 12-08-2026

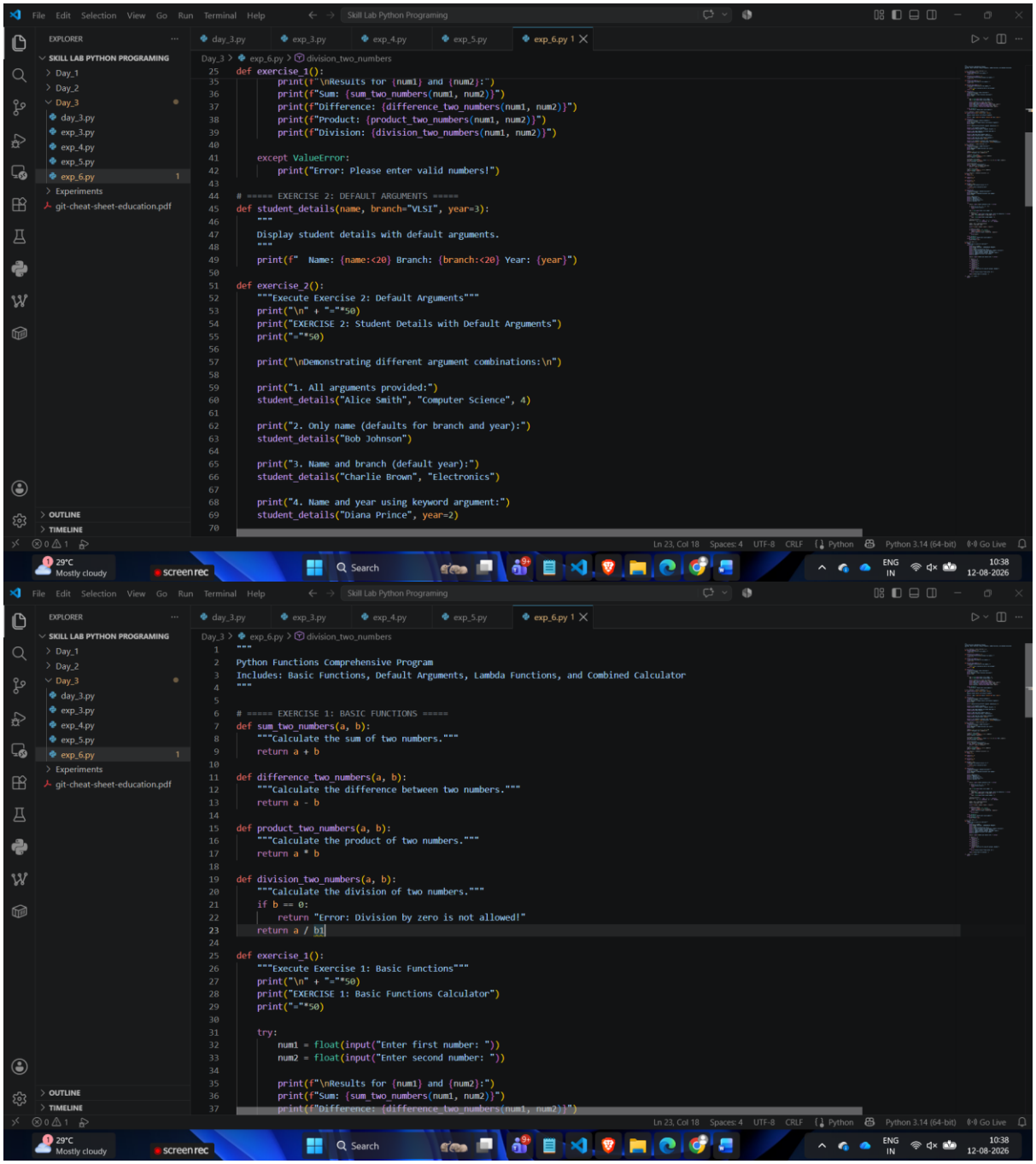
File Edit Selection View Go Run Terminal Help
Skill Lab Python Programing
EXPLORER
SKILL LAB PYTHON PROGRAMING
Day_1
Day_2
Day_3
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py
Experiments
git-cheat-sheet-education.pdf
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS G:\My Drive\Skill Lab Python Programing> python -u "g:\My Drive\Skill Lab Python Programing\Day_3\exp_6.py"
PYTHON FUNCTIONS - COMPREHENSIVE PROGRAM
Select an exercise to run:
1. Basic Functions (Sum, Difference, Product, Division)
2. Default Arguments (Student Details)
3. Lambda Functions (Squares, Even/Odd, Cubes)
4. Combined Calculator with Lambda
5. Exit
Enter your choice (1-5): 1
EXERCISE 1: Basic Functions Calculator
Enter first number: 56
Enter second number: 35
Results for 56.0 and 35.0:
Sum: 91.0
Difference: 21.0
Product: 1960.0
Division: 1.6
Press Enter to continue...
PYTHON FUNCTIONS - COMPREHENSIVE PROGRAM
Select an exercise to run:
1. Basic Functions (Sum, Difference, Product, Division)
2. Default Arguments (Student Details)
3. Lambda Functions (Squares, Even/Odd, Cubes)
4. Combined Calculator with Lambda
5. Exit
Enter your choice (1-5): 2
OUTLINE
TIMELINE
Ln 198, Col 11 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Go Live
powershell
Code
29°C Mostly cloudy
screenrec
Search
ENG IN
10:39 12-08-2026


File Edit Selection View Go Run Terminal Help
Skill Lab Python Programing
EXPLORER
SKILL LAB PYTHON PROGRAMING
Day_1
Day_2
Day_3
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py
Experiments
git-cheat-sheet-education.pdf
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py 1
Day_3 > exp_6.py > division_two_numbers
def main():
choice = input("\nEnter your choice (1-5): ").strip()
if choice == '1':
exercise_1()
elif choice == '2':
exercise_2()
elif choice == '3':
exercise_3()
elif choice == '4':
exercise_4()
elif choice == '5':
print("\nThank you for using the program. Goodbye!")
break
else:
print("Invalid choice! Please enter 1-5.")
input("\nPress Enter to continue...")
if __name__ == "__main__":
main()
OUTLINE
TIMELINE
Ln 23, Col 18
Spaces: 4
UTF-8
CRLF
Python
Python 3.14 (64-bit)
Go Live
29°C
Mostly cloudy
screenrec
Search
ENG
IN
10:39
12-08-2026

File Edit Selection View Go Run Terminal Help
Skill Lab Python Programing
EXPLORER
SKILL LAB PYTHON PROGRAMING
Day_1
Day_2
Day_3
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py
Experiments
git-cheat-sheet-education.pdf
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py 1
Day_3 > exp_6.py > division_two_numbers
def exercise_4():
# Perform operation
operations = {'1': ('+', add), '2': ('-', subtract),
'3': ('*', multiply), '4': ('/', divide)}
symbol, func = operations[choice]
result = func(num1, num2)
print(f"\n{num1} {symbol} {num2} = {result}")
# Lambda for square
if isinstance(result, (int, float)):
square = (lambda x: x ** 2)(result)
print(f"Square of result ({result})² = {square}")
else:
print(result)
except ValueError:
print("Error: Please enter valid numbers!")
except Exception as e:
print(f"Error: {e}")
# ===== MAIN MENU =====
def main():
"""Main menu to access all exercises"""
while True:
print("\n" + "="*50)
print("PYTHON FUNCTIONS - COMPREHENSIVE PROGRAM")
print("="*50)
print("\nSelect an exercise to run:")
print("1. Basic Functions (Sum, Difference, Product, Division)")
print("2. Default Arguments (Student Details)")
print("3. Lambda Functions (Squares, Even/Odd, Cubes)")
print("4. Combined Calculator with Lambda")
print("5. Exit")
choice = input("\nEnter your choice (1-5): ").strip()
OUTLINE
TIMELINE
Ln 23, Col 18
Spaces: 4
UTF-8
CRLF
Python
Python 3.14 (64-bit)
Go Live
29°C
Mostly cloudy
screenrec
Search
ENG
IN
10:39
12-08-2026


File Edit Selection View Go Run Terminal Help
Skill Lab Python Programing
EXPLORER
SKILL LAB PYTHON PROGRAMING
Day_1
Day_2
Day_3
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py
Experiments
git-cheat-sheet-education.pdf
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py 1
Day_3 > exp_6.py > division_two_numbers
def multiply(a, b):
return a * b
def divide(a, b=1):
"""Division with default divisor of 1"""
if b == 0:
return "Error: Division by zero!"
return a / b
def exercise_4():
"""Execute Exercise 4: Combined Calculator"""
print("\n" + "="*50)
print("EXERCISE 4: Advanced Calculator with Lambda")
print("="*50)
print("\nOperations:")
print("1. Addition (+)")
print("2. Subtraction (-)")
print("3. Multiplication (*)")
print("4. Division (/)")
try:
choice = input("\nEnter operation (1-4): ").strip()
if choice not in ['1', '2', '3', '4']:
print("Invalid choice!")
return
num1 = float(input("Enter first number: "))
if choice == '4':
num2_input = input("Enter second number (Enter for default=1): ").strip()
num2 = float(num2_input) if num2_input else 1
else:
num2 = float(input("Enter second number: "))
# Perform operation
OUTLINE
TIMELINE
Ln 23, Col 18 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Go Live
29°C
Mostly cloudy
screenrec
Search
ENG
IN
10:39
12-08-2026

File Edit Selection View Go Run Terminal Help
Skill Lab Python Programing
EXPLORER
SKILL LAB PYTHON PROGRAMING
Day_1
Day_2
Day_3
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py
Experiments
git-cheat-sheet-education.pdf
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py 1
Day_3 > exp_6.py > division_two_numbers
# ===== EXERCISE 3: LAMBDA FUNCTIONS =====
def exercise_3():
"""Execute Exercise 3: Lambda Functions"""
print("\n" + "="*50)
print("EXERCISE 3: Lambda Functions with Lists")
print("="*50)
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
print(f"\nOriginal list: {numbers}\n")
# Squares using lambda
squares = list(map(lambda x: x ** 2, numbers))
print(f"Squares: {squares}")
# Even/Odd using lambda
even_odd = list(map(lambda x: "Even" if x % 2 == 0 else "Odd", numbers))
print(f"Even/Odd: {even_odd}")
# Detailed display
print("\nDetailed classification:")
for num, status in zip(numbers, even_odd):
print(f" {num:2} → {status}")
# Cubes using lambda
cubes = list(map(lambda x: x ** 3, numbers))
print(f"\nCubes: {cubes}")
# ===== EXERCISE 4: COMBINED CALCULATOR =====
def add(a, b):
return a + b
def subtract(a, b):
return a - b
def multiply(a, b):
return a * b
OUTLINE
TIMELINE
Ln 23, Col 18 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Go Live
29°C
Mostly cloudy
screenrec
Search
ENG
IN
10:38
12-08-2026


File Edit Selection View Go Run Terminal Help
Skill Lab Python Programing
EXPLORER
SKILL LAB PYTHON PROGRAMING
Day_1
Day_2
Day_3
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py
Experiments
git-cheat-sheet-education.pdf
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py 1
Day_3 > exp_6.py > division_two_numbers
def exercise_1():
print(f"\nResults for {num1} and {num2}:")
print(f"Sum: {sum_two_numbers(num1, num2)}")
print(f"Difference: {difference_two_numbers(num1, num2)}")
print(f"Product: {product_two_numbers(num1, num2)}")
print(f"Division: {division_two_numbers(num1, num2)}")
except ValueError:
print("Error: Please enter valid numbers!")
# ===== EXERCISE 2: DEFAULT ARGUMENTS =====
def student_details(name, branch="VLSI", year=3):
"""
Display student details with default arguments.
"""
print(f" Name: {name:<20} Branch: {branch:<20} Year: {year}")
def exercise_2():
"""Execute Exercise 2: Default Arguments"""
print("\n" + "="*50)
print("EXERCISE 2: Student Details with Default Arguments")
print("="*50)
print("\nDemonstrating different argument combinations:\n")
print("1. All arguments provided:")
student_details("Alice Smith", "Computer Science", 4)
print("2. Only name (defaults for branch and year):")
student_details("Bob Johnson")
print("3. Name and branch (default year):")
student_details("Charlie Brown", "Electronics")
print("4. Name and year using keyword argument:")
student_details("Diana Prince", year=2)
OUTLINE
TIMELINE
Ln 23, Col 18 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Go Live
29°C Mostly cloudy
screenrec
Search
ENG IN
10:38 12-08-2026

File Edit Selection View Go Run Terminal Help
Skill Lab Python Programing
EXPLORER
SKILL LAB PYTHON PROGRAMING
Day_1
Day_2
Day_3
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py
Experiments
git-cheat-sheet-education.pdf
day_3.py
exp_3.py
exp_4.py
exp_5.py
exp_6.py 1
Day_3 > exp_6.py > division_two_numbers
"""
Python Functions Comprehensive Program
Includes: Basic Functions, Default Arguments, Lambda Functions, and Combined Calculator
"""
# ===== EXERCISE 1: BASIC FUNCTIONS =====
def sum_two_numbers(a, b):
"""Calculate the sum of two numbers."""
return a + b
def difference_two_numbers(a, b):
"""Calculate the difference between two numbers."""
return a - b
def product_two_numbers(a, b):
"""Calculate the product of two numbers."""
return a * b
def division_two_numbers(a, b):
"""Calculate the division of two numbers."""
if b == 0:
return "Error: Division by zero is not allowed!"
return a / b
def exercise_1():
"""Execute Exercise 1: Basic Functions"""
print("\n" + "="*50)
print("EXERCISE 1: Basic Functions Calculator")
print("="*50)
try:
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
print(f"\nResults for {num1} and {num2}:")
print(f"Sum: {sum_two_numbers(num1, num2)}")
print(f"Difference: {difference_two_numbers(num1, num2)}")
OUTLINE
TIMELINE
Ln 23, Col 18 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit) Go Live
29°C Mostly cloudy
screenrec
Search
ENG IN
10:38 12-08-2026